

ORIGINAL RESEARCH

Discriminating clinical prediction models: how outcome incidence and decision thresholds shape clinical utility and apt AUROC values

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Abstract

Objectives: Clinical prediction models that estimate an individual's outcome risk are often evaluated using discrimination metrics, particularly the area under the receiver-operating characteristic curve (AUROC; c-statistic). Yet what constitutes a “good” AUROC is unclear because discrimination is often interpreted without explicit reference to outcome incidence or the risk thresholds that guide clinical decisions. We aimed to show that the clinical utility associated with a given AUROC depends jointly on decision thresholds and the outcome incidence in the target population.

Methods: We conducted a simulation study and applied the findings to real examples. In simulations, we generated binary outcomes for 10,000 patients across 100 scenarios defined by combinations of target AUROC values (0.50–0.95) and overall outcome incidence (0.05–0.50), assuming perfect calibration. For each scenario, we evaluated clinical utility as net benefit across six treatment decision thresholds (5%–30%) and examined the resulting distribution of predicted risks and the proportion of patients exceeding each threshold. We then applied this framework to outcomes from a living clinical practice guideline on treatments for type 2 diabetes.

Results: The simulations and applied examples showed that, for a well-calibrated model, what constitutes a “good” AUROC depends jointly on outcome incidence and the decision thresholds used in practice. As AUROC approached 0.5, predicted risks clustered increasingly tightly around the population incidence, limiting threshold-based decision-making unless the threshold was close to the incidence (so that some patients still fell above and some below it). Higher AUROC values widened the distribution of predicted risks, but clinical utility remained contingent on the proximity of thresholds to the incidence: thresholds far from the incidence required substantially higher discrimination for patients to cross the threshold and yield benefit.

Conclusion: Whether a clinical prediction model's AUROC is “good” cannot be judged in isolation. Clinical usefulness arises from the joint relationship between discrimination, outcome incidence, and decision thresholds: For a well-calibrated model, AUROC determines the width of predicted risks, incidence anchors their center, and thresholds determine whether that width translates into actionable decisions. Under poor calibration, predicted risks may be dispersed in ways that do not reflect true risk separation, and these relationships may not

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hold. Accordingly, even well-calibrated models with high AUROC may have limited utility when thresholds are far from the outcome incidence, whereas models with modest AUROC may still be useful when thresholds lie near the incidence. Researchers should account for incidence and decision thresholds when designing prediction model studies and when interpreting the practical usefulness of a given AUROC value. © 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Keywords: Prognostic models; Discrimination; Clinical utility; Decision thresholds; Clinical practice guidelines; Risk stratification

1. Introduction

Clinical prediction models estimate an individual's risk of a future outcome by combining multiple prognostic factors (predictors). For individual treatment decisions, clinicians and patients turn to clinical prediction models to obtain patient-specific risk estimates, rather than relying solely on overall outcome incidence from a cohort. Here, we use incidence to mean the observed outcome frequency in the target population and prognosis to mean the inference that incidence supports for similar patients. For example, if a high-quality 5-year study in a defined cohort of patients with cancer reports 65% mortality (the incidence), a new patient similar to those in the cohort may receive a 5-year mortality estimate of 65% (the prognosis). Ideally, the prognosis would be more individualized, by conditioning on multiple prognostic factors that refine the incidence estimate for the individual, which is why investigators develop clinical prediction models to estimate personalized risk [1,2]. Examples include, among many others, the recently published Predicting Risk of Cardiovascular Disease EVENTS equations, designed to predict the risk of cardiovascular disease (CVD) among patients without CVD [3] and their predecessor, the pooled cohort equations [4].

A useful prediction model must outperform the average risk estimate of the outcome in the relevant population. Developers report a model's performance in terms of discrimination, calibration, and clinical utility metrics [5]. Discrimination answers a simple, if sometimes hard to implement, question: given two patients - one who experiences the outcome and one who does not - what is the probability that the model assigns the higher risk to the former? Discrimination dominates reports on evaluation of model performance in both derivation and validation studies; it is summarized by the area under the receiver-operating characteristic curve (AUROC) (equivalent to the concordance (c-) statistic) for binary outcomes and the C-index for time-to-event data [1,5,6]. A review of eight systematic reviews of 1448 prediction models found that the reporting of discrimination is increasing over time (43% in 2005/09 period, 70% in 2020/21) [7].

For sensible models, AUROC (c-statistic) ranges from 0.5 (no better than chance) to 1 (perfect separation). Historically, investigators have offered thresholds such as <0.60 "poor," 0.60–0.75 "possibly helpful," and >0.75 "useful/

good" [1], but guidance from Steyerberg emphasizes that such thresholds are context-dependent and potentially misleading [8]. Steyerberg notes it is problematic to prescribe any minimum AUROC for clinical usefulness, which depends on calibration and decision consequences, not discrimination alone [8]. Similarly, Riley et al. highlight that what counts as "high" discrimination varies by field — where strong predictors exist, 0.8 might be considered high, but in more challenging settings, values around 0.6 may be the best achievable [9]. Finally, the AUROC is sensitive to case-mix: even if the models are well-calibrated, greater patient heterogeneity can inflate AUROC, whereas a restricted spectrum can depress it [10].

Therefore, determining apt AUROC values that correspond to a clinically useful model remains challenging and potentially confusing for researchers and stakeholders. In this paper, we use simulations and real examples to show that the clinical utility associated with a given AUROC depends jointly on the outcome incidence (cumulative incidence; overall outcome probability) in the target population and the risk thresholds that guide clinical decisions. Specifically, we examine how AUROC determines the spread of predicted risks around the population incidence, and thus whether enough patients cross clinically relevant management thresholds to yield net benefit. Consistent with prior work urging decision-curve analysis rather than AUROC alone, we reframe discrimination in decision-analytic terms by showing how incidence and decision thresholds determine whether a given AUROC yields enough risk spread to support useful decisions.

2. Methods

2.1. Study overview

We conducted a stochastic simulation study in which the data-generating process was specified and therefore known. We used a factorial design crossing outcome incidence with target discrimination levels, pairing each level of one factor with every level of the other. This allowed us to quantify how outcome incidence, discriminatory ability, and decision thresholds jointly shape a model's clinical usefulness. To isolate the role of discrimination, we examined an idealized best-case setting with perfect calibration and a known data-generating mechanism; in practice, miscalibration and

What is new?**Key findings**

- Area under the receiver-operating characteristic curve (AUROC) alone does not determine clinical usefulness of prognostic models because outcome incidence and decision thresholds govern whether predicted risks cross action cutoffs and yield net benefit.
- Lower AUROC values compress predicted risks around the population incidence, whereas higher AUROC values widen risk dispersion and expand the range of decision thresholds a model can support

What this adds to what was known?

- This study reframes discrimination in decision-analytic terms by linking AUROC to the spread of predicted risks around incidence and to threshold-based decisions through net benefit and decision curves.
- This study shows, using a diabetes guideline example, why models with similar AUROC values can be fit for purpose for one outcome yet ineffective for another when the implied decision thresholds differ.

What is the implication and what should change now?

- Researchers and guideline panels should evaluate models at outcome-specific thresholds by examining predicted-risk distributions and decision curves rather than interpreting AUROC in isolation.
- Researchers and guideline panels should expect thresholds far from incidence to require substantially higher discrimination for model-guided care to outperform treat-all or treat-none strategies.

estimation error would be expected to further reduce clinical utility.

2.2. Simulation process overview

We simulated perfectly calibrated prediction settings following methods from Stahlmann et al [11]. By design, the calibration intercept (calibration-in-the-large; reflecting systematic over- or underprediction) was 0, and the calibration slope (reflecting agreement in risk spread) was 1. In other words, we examined scenarios in which the true risk model is known; we did not explore miscalibration or

finite-sample model-fitting problems, which may reduce a model's clinical utility further [12,13]. The simulation is hence an efficacy study of statistical approaches, as opposed to a pragmatic study. Simulations included the following steps.

2.2.1. Step 1 – definition of simulation parameters and scenarios

We prespecified the outcome incidence p and the target AUROC. We varied p over ten levels from 0.05 to 0.5 in increments of 0.05 and varied AUROC over ten levels from 0.50 to 0.95 in increments of 0.05, yielding 100 distinct scenario combinations. With this approach, we could see how models with poor discrimination behave in a target population with low to high incidence of an outcome and how models with superior discrimination behaves under low to high incidence of an outcome.

2.2.2. Step 2 – generating binary outcomes

For each scenario, we created a cohort of 10,000 individuals and assigned outcomes so that the marginal incidence equaled the outcome incidence p . Since the outcomes are drawn from probabilities, the number of outcomes could vary between simulation iterations. We set a random seed number for consistent results, and the resultant number of outcomes ranged from 484 under $p=0.05$ to 4957 under $p = 0.5$.

2.2.3. Step 3 – generating a continuous predictor with a prespecified AUROC

Conditional on the outcome value, we sampled each individual's value of a single continuous predictor X from normal distributions with equal variance [11]. For individuals without the outcome, a continuous predictor X_0 was sampled from a standard normal distribution $X_0 \sim N(0, 1)$. For individuals with the outcome, the probability corresponding to each AUROC was calculated with the quantile function, $\Phi^{-1}(AUROC)$, multiplied by $\sqrt{2}$. Then, these probabilities served as the mean when drawing a continuous predictor X_1 from a normal distribution $X_1 \sim N(\Phi^{-1}(AUROC) \cdot \sqrt{2}, 1)$. This ensured that a model estimating the relationship between the predictor X and the probability of an outcome would have an estimated AUROC approximate to the target AUROC that we simulated.

2.2.4. Step 4 – fit the prediction model and obtain predicted probabilities

We fitted a logistic regression model to the 10,000 simulated individuals, regressing the simulated outcome Y on the simulated predictor X . We converted the fitted linear predictor to predicted probabilities using the inverse-logit: $\hat{p} = \text{logit}^{-1}(\hat{B}_0 + \hat{B}_1 X)$. Since we generated fixed outcomes per scenario, and we evaluated performance of the model on the simulated data, predicted probabilities were

perfectly calibrated in expectation (slope 1, intercept 0). When the absolute difference between the target AUROC and the empirical AUROC computed from $\hat{p}$ exceeded 0.001, we repeated step three until the empirical AUROC fell within tolerance.

2.2.5. Step 5 – summarize discrimination and predicted probabilities distribution

For each combination of outcome incidence p and target AUROC, we plotted the distribution of predicted probabilities. For each scenario, we also computed the empirical AUROC using predicted probabilities from step 4 and outcomes simulated in step 2. These distributions illustrate how increasing AUROC widens the spread of predicted risks around the population incidence. Importantly, AUROC constrains risk dispersion but does not encode where predicted risks lie on the absolute risk scale.

2.2.6. Step 6 – analyze decision thresholds

A patient and clinician may choose to undergo a medical intervention based on the patient's predicted probability of an outcome. This choice is based on decision thresholds.

To examine the corresponding clinical utility at each threshold (p_t), we also evaluated the net benefit [14]:

$$\text{Net Benefit} = \frac{\text{True Positives}}{10,000} - \left(\frac{\text{False Positives}}{10,000} \left(\frac{p_t}{1 - p_t} \right) \right)$$

Where 10,000 is the total number of patients in each simulation and p_t is the decision threshold probability. Net benefit was calculated for each combination of outcome incidence and target AUROC. We compared the net benefit of using prediction models with treat-all and treat-none strategies in a decision curve analysis.

To ensure the same scale across different scenarios, we standardized the net benefit by the outcome incidence: $\text{Standardized Net Benefit} = \frac{\text{Net Benefit}}{\text{Simulation Outcome Incidence}}$ where Simulation outcome incidence is the proportion of patients with an outcome of 1 in a given simulation setting.

2.3. Clinical example

We operationalized the interplay between model discrimination and clinical decision threshold for two outcomes from the living clinical practice guideline that issues recommendations on medications to prevent cardiovascular and kidney complications in adults with type 2 diabetes [15]: all-cause mortality and nonfatal myocardial infarction (MI). Two evidence streams informed the example: a living network meta-analysis (NMA) that estimates relative effects for drug classes across cardiorenal outcomes [16] and a living systematic review of validated prognostic models reporting discrimination (c-statistics) and calibration for the same outcomes [17].

The guideline panel prespecified minimally important absolute risk differences (MIDs) for each outcome (eg, 1% for all-cause mortality; 2% for nonfatal MI). These MIDs serve as decision thresholds: the smallest absolute benefit typical well-informed patients would consider worthwhile. To convert from MID to the baseline risk on the risk scale, we conducted the following:

Given the NMA odds ratio (OR , denoted θ), we solved for the required baseline risk p_0 that yields the MID as an absolute risk reduction. The calculations for conversion of MID to baseline risk are provided in the supplemental material.

For each candidate model identified in the systematic review and the two outcomes, we generated predicted probability distributions consistent with (a) the chosen outcome incidence p and (b) the model's AUROC. Following the approach used in our simulations, we assumed a single latent continuous score with class-conditional normal distributions and equal variances. Using these predicted probabilities, we (i) plotted histograms to determine whether each model's spread straddles the decision threshold p_0 ; (ii) computed the proportion of patients with predicted probabilities above the threshold; and (iii) calculated net benefit and standardized net benefit, comparing each model with treat-all and treat-none strategies.

2.4. Statistical software and packages

We conducted analyses using R version 4.3.2 with the following packages: pROC for receiver operating curve analysis and AUROC calculation, ggplot2 for visualization, and dplyr and tidyr for data manipulation. We used patchwork for multipanel plot arrangement and Matrix for correlation matrix operations. Fully reproducible code can be found here: [TRCHR-CP/Discriminating-clinical-prediction-models](https://trchr-cp.github.io/Discriminating-clinical-prediction-models). We implemented the simulation workflow and figure generation in an interactive Shiny application so users can reproduce and customize the results for their own clinical context: <https://marisasignorile.shinyapps.io/DiscrimSim/>.

3. Results

We present the results as a series of key learning points. Under the perfect calibration assumption of our simulations, AUROC affects clinical usefulness indirectly: as discrimination increases, predicted risks spread farther around the population incidence, which determines how many patients cross clinically relevant decision thresholds and whether decisions yield net benefit. The figures below illustrate the same mechanism from different perspectives – AUROC determines the width of the predicted-risk distribution, incidence anchors its center, and decision thresholds determine

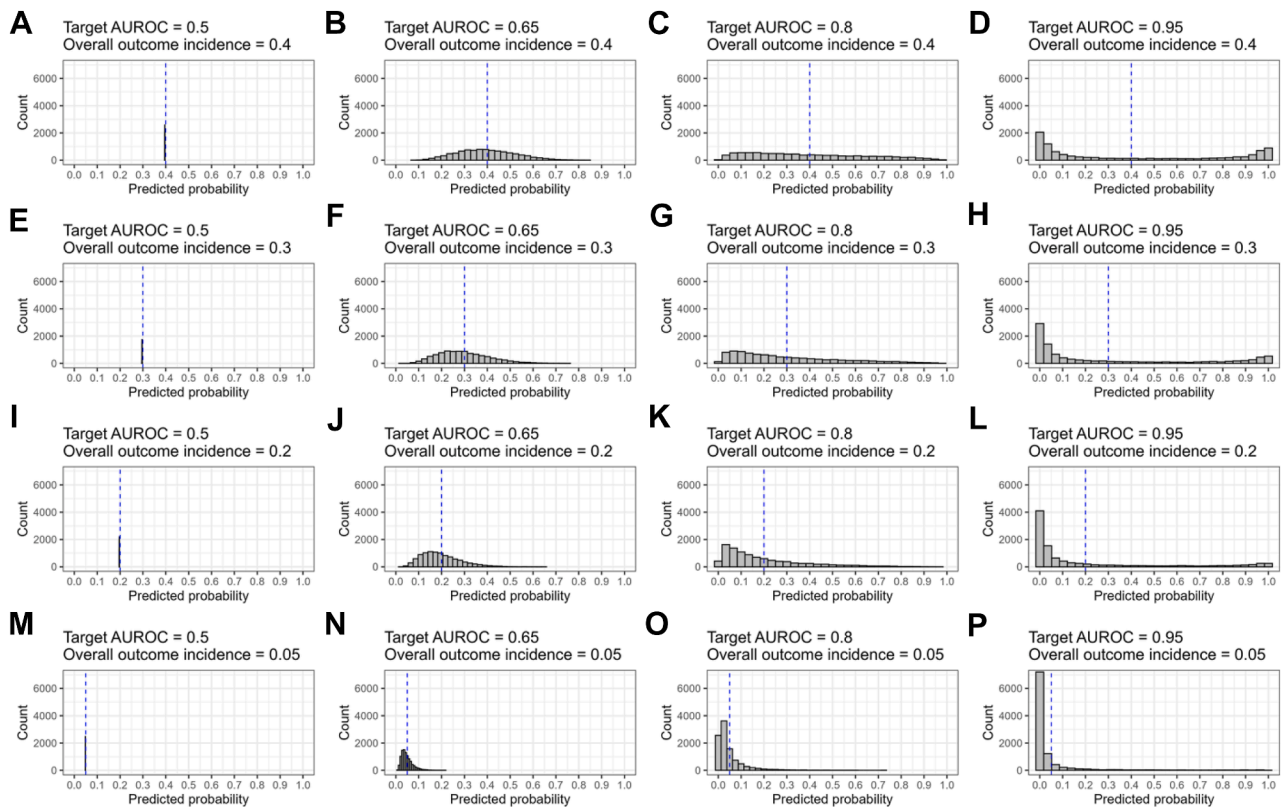


Figure 1. Histograms showing the distribution of predicted probabilities generated from simulations across different AUROC values and overall outcome incidence. Each panel corresponds to a unique AUROC/overall outcome incidence combination. Vertical blue lines indicate the overall outcome incidence for each scenario. Select combinations were included in figure based on clinical relevance. Histograms for all 100 simulations can be found in [Supplemental Figure 1](#). AUROC, area under the receiver-operating characteristic curve. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article).

whether that spread translates into different treatment decisions.

3.1. Lower AUROC values correspond to narrower spread of estimated outcome probabilities

Histograms show how lower discrimination constrains the distribution of predicted probabilities ([Fig 1](#) and [Supplemental Fig 1](#)). When AUROC = 0.5 (far left column), the model assigns nearly the same probability to everyone. Under perfect calibration, that single value equals the cohort's overall outcome incidence, so predicted risks show essentially no spread. In that setting, the model adds no clinically useful information beyond the population incidence. As AUROC increases, predicted probabilities widen around the incidence: some patients move toward very low risks, while others move toward much higher risks. This widening allows decision thresholds to separate patients into groups above vs. below the threshold. [Figure 1](#) therefore illustrates the core mechanism underlying all subsequent results: discrimination determines how widely predicted risks can spread around the population incidence, and without sufficient spread, threshold-based decision-making cannot yield benefit.

3.2. The gap between decision threshold and overall outcome incidence governs utility; higher AUROC widens the useable range

In [Figure 2](#) (and [Supplementary Fig 2](#)), we plot standardized net benefit across decision thresholds for each outcome-incidence level, with separate curves by AUROC. A consistent pattern emerges: models yield positive standardized net benefit when the chosen decision threshold lies close enough to the population incidence for the model's risk spread to place some patients above and some below the threshold. As the gap between threshold and incidence widens, predicted risks from low-AUROC models cluster tightly around the incidence, few patients cross the threshold, and net benefit approaches the treat-all or treat-none baselines. Increasing AUROC widens the predicted-risk distribution and therefore expands the range of decision thresholds that the model can support, but it does not guarantee utility at thresholds far from the incidence.

Notably, this useable threshold window also depends on the outcome incidence itself. As incidence decreases, the predicted-risk distribution anchors closer to the lower bound of the probability scale, compressing the

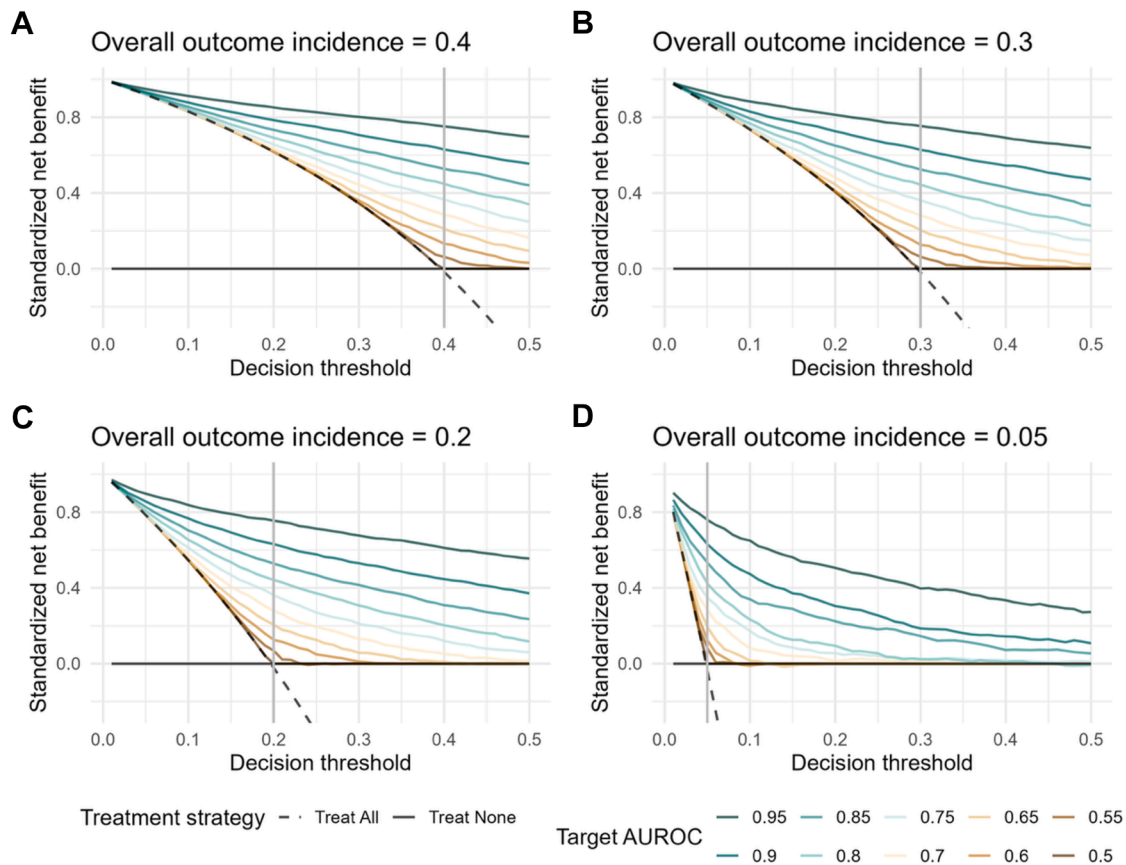


Figure 2. Decision curve analyses illustrating the standardized net benefit across varying AUROC targets, under outcome incidences of 0.4, 0.3, 0.2, and 0.1. The vertical line indicates the moment the threshold equals the outcome incidence. The closer thresholds are to this line, the higher net benefit of prediction models, regardless of AUROC. AUROC, area under the receiver-operating characteristic curve.

absolute range over which risks can spread. Consequently, at lower incidence levels (eg, 0.05), the window of thresholds where the model outperforms treat-all and treat-none is narrower than at higher incidence levels (eg, 0.40), further underscoring that incidence is not merely the anchor of predicted risks but also a determinant of how much room exists for threshold-based decisions to confer utility.

High-incidence settings illustrate “treat-all pressure.” At low decision thresholds when outcome incidence is high, curves for lower AUROC track the treat-all line because most patients exceed the threshold. Higher AUROC moves the model’s curve away from treat-all, but gains remain modest until AUROC becomes large enough to identify a meaningful subgroup below the threshold. In low-incidence settings at high thresholds, curves approach the treat-none line unless AUROC is high enough to place some patients above the threshold.

The edges of the threshold axis show unavoidable convergence. As thresholds approach 0 or 1, standardized

net benefit compresses toward the treat-all and treat-none baselines, regardless of AUROC. In practical terms, extremely permissive or extremely conservative thresholds leave little room for any model to improve decisions.

Figure 3 reframes these results as differences in standardized net benefit vs. treat-all and treat-none. The two difference curves intersect at the threshold equal to the incidence, where treat-all and treat-none yield the same standardized net benefit and the model’s advantage is symmetric around that point. To the left of the intersection in high-incidence panels, treat-all tends to dominate unless discrimination is high; to the right of the intersection in low-incidence panels, treat-none tends to dominate unless discrimination is high. The area under the cross represents the range of decision thresholds a model can support given its discrimination: it remains minimal at AUROC = 0.55 and increases as AUROC rises. In summary, utility does not arise at fixed AUROC values; it arises when the decision threshold lies sufficiently close to the incidence for the model’s risk spread to move patients across that threshold.

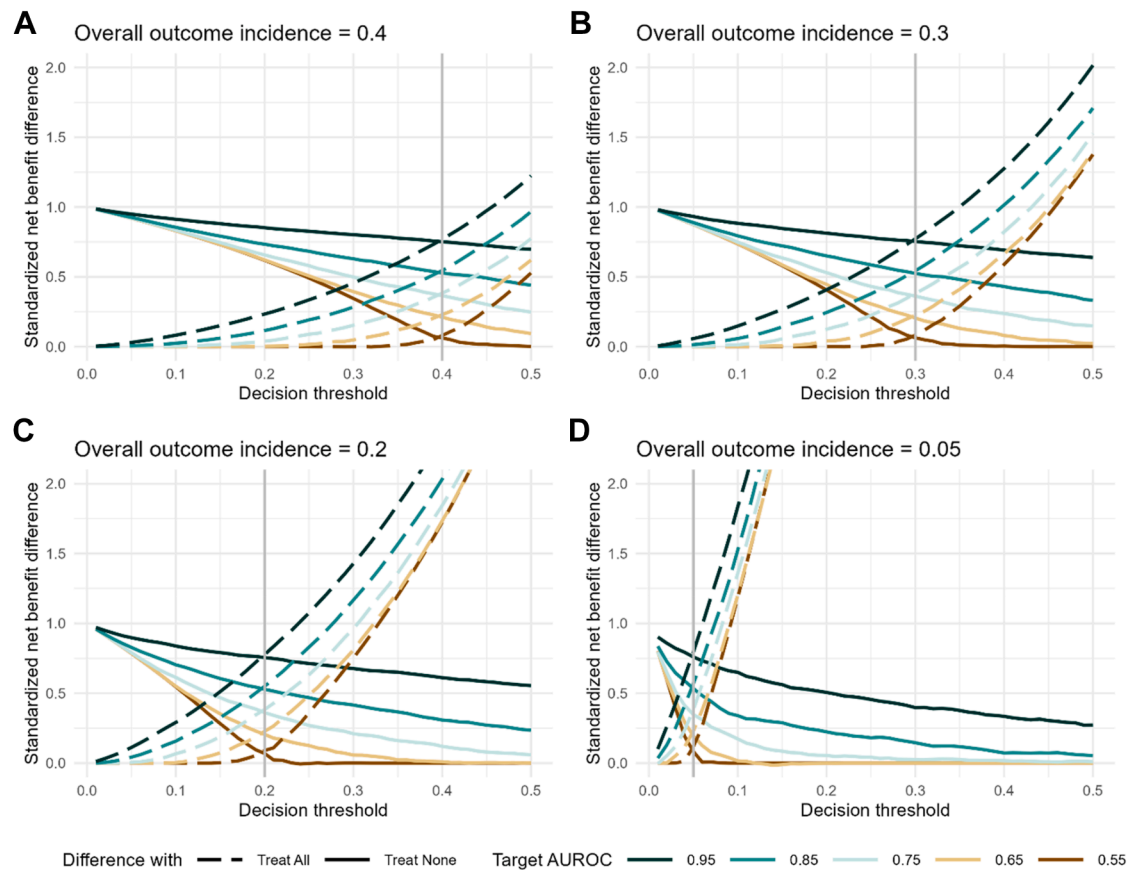


Figure 3. Decision curve analyses showing the difference in standardized net benefit compared to treating no patients and treating all patients, across varying discrimination thresholds, under outcome incidences of 0.5, 0.4, 0.2, and 0.1. When incidences are low, the steep slopes of prediction models quickly reach higher net benefit than treat-all, regardless of AUROC. When incidences are high, the difference in net benefit tends toward zero, with clear variation between AUROC levels. Vice versa for treat-none. AUROC, area under the receiver-operating characteristic curve.

3.3. Decision thresholds and outcome incidence determine what AUROC values are apt: diabetes drugs example

We illustrate how outcome incidence and decision thresholds determine what AUROC values are apt for clinical utility. A BMJ rapid recommendation panel developed recommendations for medications in adults with type 2 diabetes [15]. Two linked reviews supported the work: a living NMA of drug effects across cardiovascular and kidney outcomes [16] and a living review of validated prognostic models for the same outcomes [17]. The panel prespecified MIDs: 1% over 5 years for all-cause and 2% for nonfatal MI. These MIDs imply baseline-risk thresholds: clinicians need an individualized baseline-risk estimate from a prognostic model to determine whether a given patient's risk is high enough that treatment would deliver at least the MID.

3.3.1. All-cause mortality (MID close to incidence): moderate discrimination can suffice

For all-cause mortality, the NMA pooled 263 trial comparisons (357,369 participants; 15,703 deaths) and found

that sodium-glucose transport protein-2 inhibitors reduce all-cause mortality vs. standard treatment (odds ratio [OR] 0.88, 95% CI 0.83–0.94). The overall event rate in the trials was ~4.1%. Using the panel's MID (1% absolute reduction over 5 years) and the OR, our calculation gives a baseline-risk threshold of approximately 9.1%. This threshold lies close to the outcome incidence (~4.1%), so a model does not need extremely high discrimination to place some patients above and some below the action threshold. In the living prognostic-model review, discrimination for all-cause mortality ranged from roughly 0.73 to 0.75, with Risk Equations for Complications Of type 2 Diabetes supported as having acceptable discrimination and calibration. Those models appear fit-for-purpose here (Fig 4), although calibration still warrants attention. Here, the decision threshold lies close to the population incidence, so even models with moderate discrimination can meaningfully separate patients around the action threshold.

3.3.2. Nonfatal MI (threshold far into the upper tail): high discrimination becomes necessary

For nonfatal MI, the NMA included > 300,000 participants and ~9000 nonfatal MIs, with an overall event rate of about

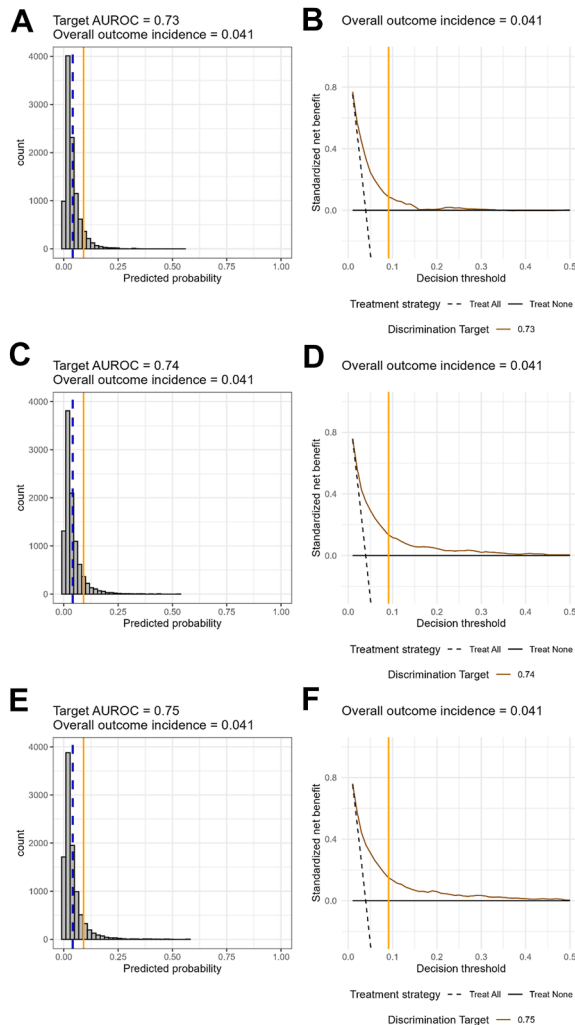


Figure 4. Histograms of predicted probability and standardized net benefit plots based on real world studies of all-cause mortality. Orange lines represent the required decision threshold baseline risk of 9.1%. AUROC, area under the receiver-operating characteristic curve. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

2.9% over 5 years. glucagon-like peptide-1 receptor agonists and sodium-glucose transport protein-2 inhibitors show modest relative effects on nonfatal MI (eg, OR ~ 0.90 vs. standard treatment). The panel's MID for nonfatal MI is an absolute 2% reduction over 5 years; plugging these values into our calculation yields a baseline-risk threshold of 26.5%. To achieve a 2% absolute reduction when OR ≈ 0.90 , a patient must have a 5-year MI risk of $\sim 26.5\%$, which only a small upper tail of the population will exceed. Identifying those patients reliably requires a more discriminating model than is typical for MI: models with c-statistics around 0.64–0.72 (eg, United Kingdom Prospective Diabetes Study-Outcomes Model 2 [UKPDS-OM2], Risk Equations for Complications Of type 2 Diabetes for MI) will often fail to create enough separation near the 25%–30% region (Fig 5). The challenge here is not that MI models perform poorly in absolute terms, but that the

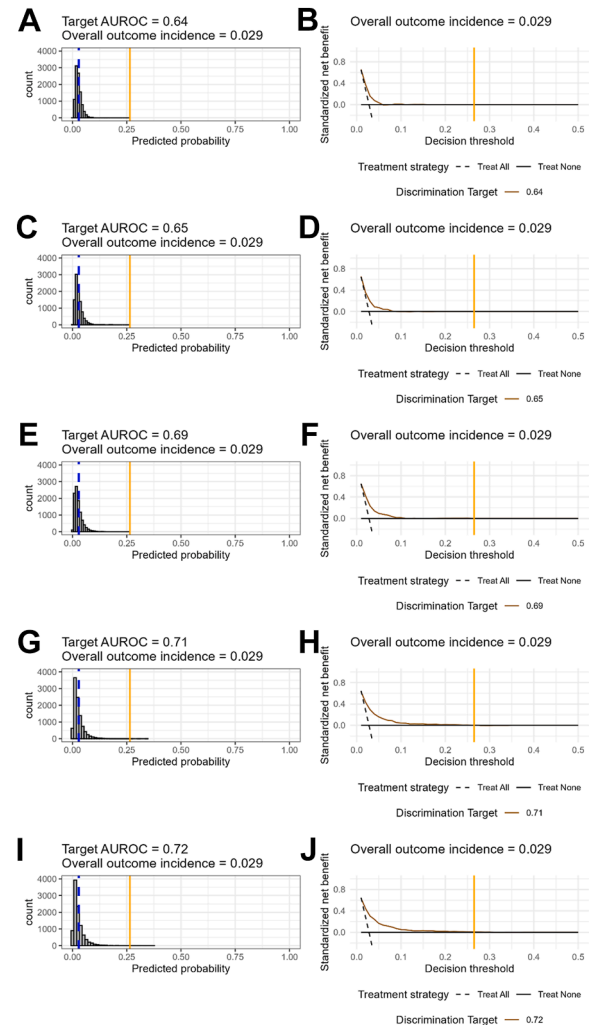


Figure 5. Histograms of predicted probability and standardized net benefit plots based on real world studies of nonfatal MI. Orange lines represent the required decision threshold baseline risk of 26.5%. AUROC, area under the receiver-operating characteristic curve. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

MID implies a decision threshold far from the incidence, deep in the upper tail of the risk distribution.

Taken together, these examples reinforce the central framework: discrimination, incidence, and decision thresholds jointly determine clinical usefulness — AUROC controls the width of the risk distribution, incidence anchors its center, and thresholds determine whether that width translates into actionable decisions.

4. Discussion

This study shows that clinicians and researchers cannot judge whether an AUROC is “good” in isolation. Discrimination affects clinical usefulness only indirectly: AUROC determines how widely predicted risks spread around the population incidence, and decision thresholds determine

whether spread moves enough patients across the action cutoff to yield net benefit. We therefore reframe discrimination in decision-analytic terms by linking AUROC, incidence, and thresholds to the consequences of threshold-based decisions.

We demonstrated this mechanism in simulations that isolated discrimination under idealized, best-case conditions (perfect calibration and a known data-generating mechanism). In this setting, lower AUROC values forced predicted risks to cluster tightly around the outcome incidence, leaving little room for threshold-based decisions unless thresholds lay near the incidence. Higher AUROC values widened the predicted-risk distribution, but thresholds far from the incidence still required substantially higher discrimination before enough patients crossed the threshold to generate benefit. These best-case assumptions strengthen the main message: miscalibration, estimation error, and model misspecification in practice would typically reduce clinical utility further, not increase it [12,13].

Applying this framework to a living guideline on medications for type 2 diabetes illustrates why models can appear similar by AUROC yet differ in usefulness for specific decisions [15–17]. When the implied decision threshold lies near the population incidence (eg, all-cause mortality), models with modest discrimination and acceptable calibration can still move a meaningful fraction of patients across the clinical threshold and yield net benefit. When the implied decision threshold lies far into the upper tail (eg, nonfatal MI), only highly discriminating models will classify enough patients above the action threshold to change treatment decisions.

The guideline encouraged risk-stratified care and pointed to prognostic models as implementation tools, but it did not distinguish, at each outcome-specific threshold, which available models likely support the intended decision vs. which models remain ineffectual. This omission can produce heterogeneous implementation: some clinicians may apply any model with a “reasonable” AUROC even when its predicted-risk spread cannot move patients across the relevant threshold, while others may default to treat-all or treat-none strategies even when a fit-for-purpose model exists.

Our framework closes this gap by coupling incidence, thresholds, and discrimination to define practical usefulness for a specific decision. For any intended decision, analysts should (i) state the population’s outcome incidence (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis+Artificial Intelligence [TRIPOD-AI] items 20a, 20b), (ii) state the clinical threshold – or plausible range of thresholds – anchored in patient-important differences, recognizing that thresholds may vary across clinician and patient preferences (TRIPOD+AI items 3b, 15), (iii) examine the predicted-risk distribution to quantify how many patients fall above and below the threshold (TRIPOD+AI items 12e, 23a), and (iv) report net benefit and decision curves at that

threshold or across a clinically plausible range to compare model-guided care with treat-all and treat-none (TRIPOD+AI items 12e, 23a) [18,19].

A model becomes useful when its predicted-risk spread plausibly straddles the threshold and yields positive (standardized) net benefit; it becomes useless when it fails to move patients across the threshold or when treat-all or treat-none dominates. Calibration remains essential: miscalibration can misclassify patients around the threshold and further reduce clinical utility [12,13].

This perspective also clarifies how guideline panels and researchers should evaluate models for risk-stratified recommendations. Instead of asking whether a model’s AUROC is “good,” panels should ask whether calibration is acceptable and whether discrimination suffices at the intended threshold given the outcome incidence. Existing frameworks help panels set minimally important differences and thresholds [20], but few approaches formally integrate incidence, thresholds, and model performance into risk-stratified recommendations. Our results argue for making incidence an explicit first step and for using decision curves to show where model-guided care improves on treat-all and treat-none—shifting attention from AUROC alone toward threshold-relevant usefulness and clinical utility [19].

Our findings also have implications for how systematic reviews evaluate and synthesize clinical prediction models. Reviews that filter or rank candidate models primarily by AUROC thresholds – for example, excluding models below 0.7 as “poor” – may inadvertently discard models that are clinically useful when the decision threshold lies near the outcome incidence, while elevating models whose discrimination, though nominally high, does not translate into utility at the intended threshold. We encourage systematic reviewers to move beyond AUROC-based rankings and instead assess whether each model’s predicted-risk spread supports threshold-based decisions for the clinical question under review. We recognize that shifting long-standing cultural norms around AUROC interpretation will require sustained effort across the prediction modeling community, but reframing model evaluation in decision-analytic terms offers a concrete starting point.

5. Conclusion

Whether a clinical prediction model’s AUROC is “good” depends on the decision thresholds used in practice, the outcome incidence in the target population, and the calibration of predicted risks. Even well-calibrated models with high AUROC may have limited clinical utility when thresholds lie far from the outcome incidence; conversely, models with modest AUROC may still have clinical utility when thresholds lie near the incidence. Ultimately, clinical usefulness arises from the joint relationship between discrimination, incidence, and thresholds:

discrimination determines the width of predicted risks, incidence anchors their center, and thresholds determine whether that width translates into actionable decisions. Researchers should incorporate incidence, thresholds, and calibration when designing prediction model studies and when interpreting AUROC for practice.

CRedit authorship contribution statement

Farid Foroutan: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jessica Weiss:** Writing – review & editing, Visualization, Methodology, Formal analysis. **Martin Mayer:** Writing – review & editing, Validation, Methodology. **Lena Kristin Bache-Mathiesen:** Writing – review & editing, Visualization, Validation. **Thomas Agoritsas:** Writing – review & editing, Validation, Methodology. **Per Olav Vandvik:** Writing – review & editing, Visualization, Investigation. **Marisa Signorile:** Validation, Software. **Joie Ensor:** Writing – review & editing, Methodology. **Arnav Agarwal:** Writing – review & editing, Validation. **Gordon Guyatt:** Writing – review & editing, Supervision, Conceptualization. **Richard D. Riley:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

Richard Riley receives royalties for sales of his textbooks “Prognosis Research in Healthcare” and “Individual Participant Data Meta-analysis.”

Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclinepi.2026.112293>.

Data availability

I have shared the link to the code for data generation process.

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